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Combined comparison: four moment closures vs Gillespie

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Problem statement

This vignette puts the four node-centric SIS approximations now shipped by `NodeBasedModels.jl` on one common benchmark:

1. Standard pairwise: Keeling triple closure, no reinfection lift.
2. Reinfection counting: Approximation 1 of Keeling et al. (2016), with the infection-history cap set to $L = 1$.
3. Motif closure: Approximation 2, using $m = 3$ motifs.
4. Neighbourhood model: Approximation 3, using the implemented $n = 2$ ego-neighbourhood closure.
5. Gillespie SSA: a seeded stochastic ensemble on the same random 3-regular host.

The disease is SIS on a $k = 3$ regular graph with $\beta = 0.5$, $\gamma = 0.25$, $N = 1000$, initial prevalence $\epsilon = 0.01$, and horizon $t_{\max} = 120$. These derive from the canonical SIR parameters ($\gamma = 0.25$, anchored to $R_0 = 2$ via the homogeneous pairwise formula $R_0 = \tau(k - 2)/\gamma$ for $k = 3$).

For the four ODE approximations below we pass no explicit **reltol/abstol**. The calls therefore exercise the package defaults: `reltol = 1e-8`, `abstol = 1e-10`.

Setup

```
ENV["GKSwstype"] = "100" # offscreen GR output for reproducible Quarto renders
using NodeBasedModels
using Graphs
using Plots
using Random
using StableRNGs
using Statistics
using Printf
using Markdown
```

`DataFrames.jl` and `PrettyTables.jl` are not dependencies of this package environment, so the summary table is generated directly as Markdown.

[!NOTE]

$R_0 = 2$ anchor. This homogeneous $k = 3$ SIS benchmark uses the `NodeBasedModels` pairwise formula $R_0 = \tau(k - 2)/\gamma$. With $\gamma = 0.25$, this gives $\tau = 0.5$ and 1% initial infection.

```

N          = 1_000
k          = 3
γ_val      = 0.25
R0_target  = 2.0
β_val      = R0_target * γ_val / (k - 2) # homogeneous pairwise anchor
ε_val      = 0.01
tmax       = 120.0
ensemble   = 48
save_dt    = 1.0

tgrid = collect(0.0:save_dt:tmax)
initial_infected = collect(1:round(Int, ε_val * N))

sis = sis_model(τ = :β)
hom = regular_network(k)

```

```
HomogeneousNetwork(3, 0.0, 1.0)
```

Shared host and Gillespie ground truth

The host graph uses the same stable seed as vignette 12 and the neighbourhood comparison test. The stochastic ensemble uses explicit per-run seeds so the mean and one-standard-deviation ribbon are reproducible across renders.

```

rng_host = StableRNG(20240301)
g         = random_regular_graph(N, k; rng = rng_host)
net       = GraphNetwork(g)

n_triangles = sum(triangles(g)) ÷ 3
n_p3_count  = sum((length(neighbors(g, v)) *
                    (length(neighbors(g, v)) - 1)) ÷ 2
                  for v in 1:nv(g)) - 3 * n_triangles

@printf("Host graph: N=%d, k=%d, triangles=%d, P3=%d\n",
        N, k, n_triangles, n_p3_count)

```

```
Host graph: N=1000, k=3, triangles=2, P3=2994
```

```

run_prev = zeros(ensemble, length(tgrid))
for r in 1:ensemble
    res = gillespie_sis(net;
                        infection_rate =  $\beta$ _val,
                        recovery_rate =  $\gamma$ _val,
                        initial_infected = initial_infected,
                        tmax = tmax,
                        seed = 20240301 + r)
    for (i, t) in enumerate(tgrid)
        run_prev[r, i] = count(res(t)) / N
    end
end

gill_prev = vec(mean(run_prev; dims = 1))
gill_sd = vec(std(run_prev; dims = 1))
@printf("Gillespie mean at t=%.1f: %.5f  $\pm$  %.5f (1 $\sigma$ , n=%d)\n",
        tmax, gill_prev[end], gill_sd[end], ensemble)

```

Gillespie mean at t=120.0: 0.81846 $\pm$ 0.01438 (1 σ , n=48)

Four ODE integrations

1. Standard pairwise: Keeling closure, no reinfection lift

```

psys_pair = generate_pairwise(sis, hom, KeelingClosure();
                             tspan = (0.0, tmax),
                             seed_fraction =  $\epsilon$ _val)
sol_pair = solve_pairwise(psys_pair, Dict(: $\beta$   $\Rightarrow$   $\beta$ _val, : $\gamma$   $\Rightarrow$   $\gamma$ _val);
                        saveat = save_dt)
I_pair = sol_pair[psys_pair.singles[:I]]

```

```

121-element Vector{Float64}:
 0.01
 0.023230379896092092
 0.04091090457319238
 0.06730779538628143
 0.1074645292486584
 0.1670520897662165
 0.2507487836959342

```

```

0.3585348767837066
0.48109973485059493
0.5994073581877432
?
0.8181818181818182
0.8181818181818185
0.8181818181818182
0.8181818181818181
0.8181818181818179
0.8181818181818178
0.818181818181818
0.8181818181818181
0.8181818181818181

```

2. Reinfection counting: $L = 1$

The lifted system uses the same Keeling closure as the standard pairwise run. Because `regular_network(k)` has clustering coefficient $\phi = 0$, Keeling's formula reduces algebraically to the Bernoulli/ordinary pair closure here; using the same closure keeps the comparison focused on the reinfection-count lift.

```

psys_re = generate_pairwise(with_reinfection_counting(sis, 1),
                           hom, KeelingClosure();
                           tspan = (0.0, tmax),
                           seed_fraction = ε_val)
sol_re   = solve_pairwise(psys_re, Dict(:β ⇒ β_val, :γ ⇒ γ_val);
                           saveat = save_dt)
I_re     = reinfection_totals(psys_re, sol_re)[:I]

```

121-element Vector{Float64}:

```

0.01
0.0231882771585825
0.04014534331168273
0.06348110900584053
0.09577373356975352
0.139716747943424
0.19772492243183887
0.2710401688386217
0.35841731019452044
0.4549037421119854
?

```

```

0.8181818181818138
0.8181818181818139
0.8181818181818143
0.8181818181818153
0.8181818181818163
0.8181818181818165
0.8181818181818163
0.8181818181818162
0.8181818181818163

```

3. Motif closure: $m = 3$

The motif comparison intentionally stops at $m = 3$. Vignette 11 and `test/runtests.jl` lines 2107–2125 document why the apparent next step ($m = 4$ on a random 3-regular host) is not an honest monotone refinement: the order-4 Kirkwood closure is blocked by the Lean-certified marginalisation obstruction `EBCMCategory.MarginalisationCharacterization.kirkwood_form_not_equivariant (T3b)` in `EdgeBasedModels.jl/proofs/EBCMCategory/MarginalisationCharacterization.lean`. The test suite records the corresponding empirical expectations as two `@test_broken` assertions rather than relaxing them away.

```

sys_motif = motif_based_sis( $\beta$  =  $\beta\_val$ ,  $\gamma$  =  $\gamma\_val$ ,  $k$  =  $k$ ,  $m$  = 3,
                           tspan = (0.0, tmax),
                           N = Float64(N),  $\varepsilon$  =  $\varepsilon\_val$ ,
                           n_p3 = Float64(n_p3_count),
                           n_c3 = Float64(n_triangles))
sol_motif = solve_motif(sys_motif; saveat = save_dt)
iI_motif = sys_motif.index[:, :I])
I_motif = [u[iI_motif] / N for u in sol_motif.u]

```

121-element Vector{Float64}:

```

0.01
0.023194206765105915
0.04031783808008257
0.0644912306590342
0.09923987612545633
0.14854146310458355
0.21594760092670387
0.3026541702506041
0.40487172819789946
0.5123511850681991

```

```

0.8162801123637278
0.8162801123637289
0.8162801123637293
0.8162801123637292
0.8162801123637287
0.8162801123637279
0.8162801123637269
0.8162801123637261
0.8162801123637256

```

4. Neighbourhood closure: $n = 2$

```

sys_nbr = generate_neighbourhood(sis, k, 2;
                                 $\beta$  =  $\beta\_val$ ,  $\gamma$  =  $\gamma\_val$ ,
                                N = 1.0,  $\epsilon$  =  $\epsilon\_val$ ,
                                tspan = (0.0, tmax))
sol_nbr = solve_neighbourhood(sys_nbr; saveat = save_dt)
I_nbr    = neighbourhood_compartment(sys_nbr, sol_nbr, :I)

```

121-element Vector{Float64}:

```

0.01
0.023196223638221682
0.04023851308048399
0.06384666982502725
0.0968258744898807
0.14228565930026962
0.20315327340767572
0.2809895273596011
0.37419971092605103
0.4764196364530713
␣
0.8173494495082438
0.8173494495082436
0.8173494495082434
0.8173494495082432
0.817349449508243
0.8173494495082427
0.8173494495082426
0.8173494495082426
0.8173494495082425

```

All ODE outputs are saved on the same grid as the Gillespie ensemble.

```
@assert collect(sol_pair.t) == tgrid
@assert collect(sol_re.t)   == tgrid
@assert collect(sol_motif.t) == tgrid
@assert collect(sol_nbr.t)  == tgrid
```

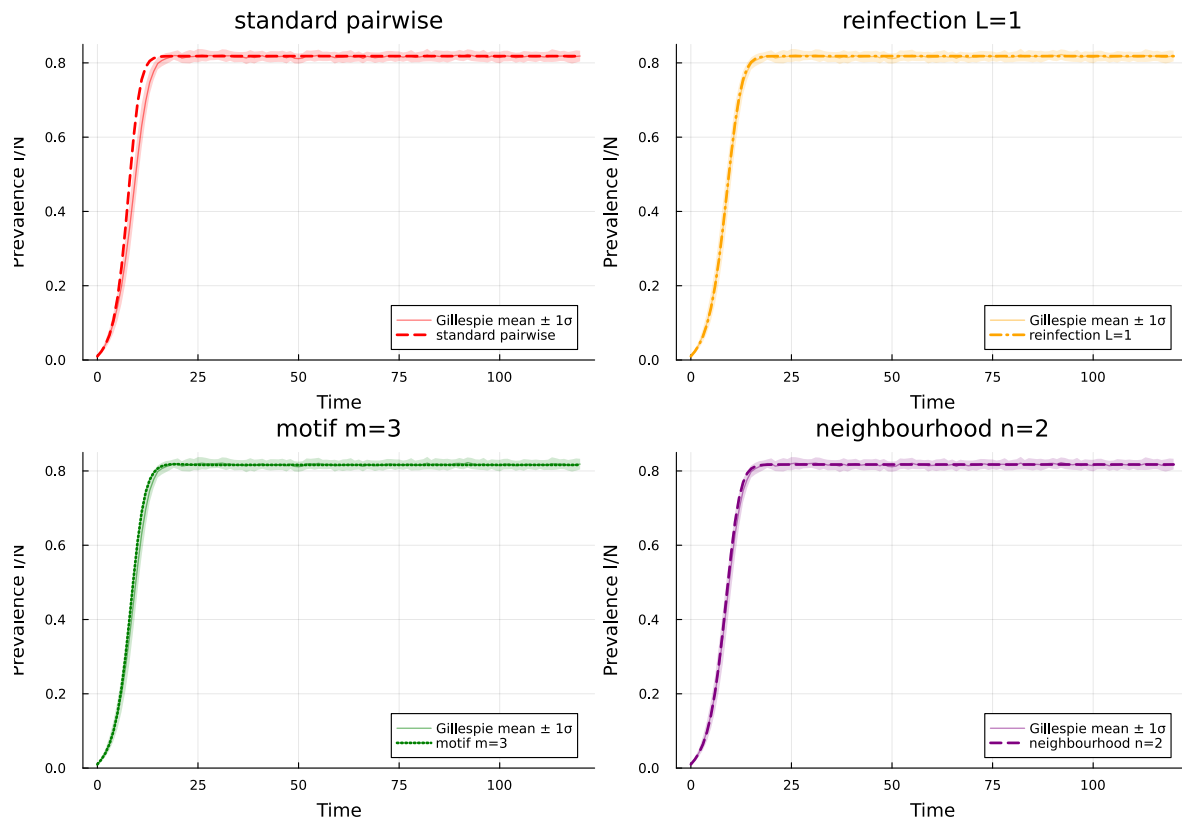
Per-method comparison panels

```
function comparison_panel(y, title, color, linestyle)
    p = plot(tgrid, gill_prev;
            ribbon = gill_sd,
            fillalpha = 0.18,
            linealpha = 0.5,
            lw = 1.2,
            color = color,
            label = "Gillespie mean  $\pm 1\sigma$ ",
            xlabel = "Time",
            ylabel = "Prevalence I/N",
            title = title,
            legend = :bottomright,
            ylims = (0.0, 0.85))
    plot!(p, tgrid, y; lw = 2.4, color = color, ls = linestyle,
          label = title)
    return p
end

panels = [
    comparison_panel(I_pair, "standard pairwise", :red, :dash),
    comparison_panel(I_re, "reinfection L=1", :orange, :dashdot),
    comparison_panel(I_motif, "motif m=3", :green, :dot),
    comparison_panel(I_nbr, "neighbourhood n=2", :purple, :dash),
]

plot(panels...; layout = (2, 2), size = (1050, 760),
     plot_title = "Each node-based approximation over Gillespie")
```


Each node-based approximation over Gillespie



Endpoint summary

```
results = [
    (name = "Gillespie mean", endpoint = gill_prev[end], signed = 0.0,
      absdev = 0.0),
    (name = "Standard pairwise (Keeling)", endpoint = I_pair[end],
      signed = I_pair[end] - gill_prev[end],
      absdev = abs(I_pair[end] - gill_prev[end])),
    (name = "Reinfection counting L=1", endpoint = I_re[end],
      signed = I_re[end] - gill_prev[end],
      absdev = abs(I_re[end] - gill_prev[end])),
    (name = "Motif m=3", endpoint = I_motif[end],
      signed = I_motif[end] - gill_prev[end],
      absdev = abs(I_motif[end] - gill_prev[end])),
    (name = "Neighbourhood n=2", endpoint = I_nbr[end],
      signed = I_nbr[end] - gill_prev[end],
```

```

        absdev = abs(I_nbr[end] - gill_prev[end])),
    ]

    lines = String[
        "| Method | Endpoint I/N | Signed deviation | Absolute deviation |",
        "|---|---:|---:|---:|",
    ]
    for r in results
        push!(lines, @sprintf("| %s | %.5f | %+.5f | %.5f |",
                               r.name, r.endpoint, r.signed, r.absdev))
    end
    display(Markdown.parse(join(lines, "\n")))

```

Method	Endpoint I/N	Signed deviation	Absolute deviation
Gillespie mean	0.81846	+0.00000	0.00000
Standard pairwise (Keeling)	0.81818	-0.00028	0.00028
Reinfection counting $L=1$	0.81818	-0.00028	0.00028
Motif $m=3$	0.81628	-0.00218	0.00218
Neighbourhood $n=2$	0.81735	-0.00111	0.00111

At this parameter point, all four deterministic approximations lie well inside the endpoint sampling spread of the 48-run Gillespie ensemble ($1\sigma = \{\text{ julia } @sprintf("%.5f", gill_sd[end])\}$). The standard pairwise and reinfection-counting endpoints are essentially tied here; motif $m = 3$ and neighbourhood $n = 2$ sit slightly below the stochastic mean. That is the honest outcome for this seeded endemic-prevalence benchmark, not a universal ranking.

Transient diagnostic

The endpoint hides the reinfection-counting effect because the lifted infection-count classes saturate at $p = L$ by the endemic plateau. A max-over-time diagnostic shows that the transient is not a no-op.

```

function max_abs_delta(a, b)
    δ = abs.(a .- b)
    i = argmax(δ)
    return (value = δ[i], time = tgrid[i])
end

transient_rows = [

```

```

("standard pairwise vs reinfection L=1", max_abs_delta(I_pair, I_re)),
("standard pairwise vs Gillespie mean", max_abs_delta(I_pair, gill_prev)),
("reinfection L=1 vs Gillespie mean", max_abs_delta(I_re, gill_prev)),
]

lines = String[
  "| Comparison | Max absolute difference | Time of max |",
  "|---|---:|---:|",
]
for (name, d) in transient_rows
  push!(lines, @sprintf("| %s | %.5f | %.1f |", name, d.value, d.time))
end
display(Markdown.parse(join(lines, "\n")))

```

Comparison	Max absolute difference	Time of max
standard pairwise vs reinfection L=1	0.14450	9.0
standard pairwise vs Gillespie mean	0.15074	9.0
reinfection L=1 vs Gillespie mean	0.01140	12.0

Honest framing

Vignette 12 already found that reinfection counting with $L = 1$, motif closure with $m = 3$, and the neighbourhood model with $n = 2$ can be nearly indistinguishable on the random 3-regular SIS benchmark. This render repeats that lesson with the standard pairwise curve added. The ordinary pairwise closure is expected to be the least reliable when infection-history heterogeneity or ego-neighbourhood correlations are the scientific target; the transient diagnostic is the appropriate place to see the reinfection-counting correction on this benchmark, while the endemic endpoint alone is too coarse to force that ranking.

The absence of motif $m = 4$ is deliberate. The Lean file `EdgeBasedModels.jl/proofs/EBCMCategory/Marginalis` states T3b as `kirkwood_form_not_equivariant`: a nontrivial multiplicative Kirkwood-form closure cannot, in general, commute with a surjective linear marginalisation. In practical terms, the $m = 4$ Kirkwood RHS does not have to marginalise to a better $m = 3$ RHS, and on the package's random 3-regular comparison it does not. The corresponding `@test_broken` assertions in `test/runtests.jl` lines 2124–2125 are therefore documentation of a certified obstruction, not unfinished implementation work.

Reproducing this vignette

- Host graph seed: `StableRNG(20240301)`.

- Gillespie ensemble: 48 runs, seeds 20240302:20240349.
- Initial infected set: `collect(1:10)` (1% of $N = 1000$).
- ODE solver tolerances: package defaults, currently `reltol = 1e-8` and `abstol = 1e-10` for `solve_pairwise`, `solve_motif`, and `solve_neighbourhood`.
- Package versions are those in `NodeBasedModels.jl/Project.toml` and the active Julia environment used by Quarto.

```
@printf("Julia %s\n", string(VERSION))
@printf("NodeBasedModels path: %s\n", pathof(NodeBasedModels))
```

Julia 1.12.5

NodeBasedModels path: /Users/sdwfrost/Projects/edgebasedmodels/NodeBasedModels.jl/src/Node

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use [NetworkOutbreaks.jl](#)'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette [01_sir_on_graphs](#) for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.